

Plant Disease Knowledge Base

Structured Reference for AI-Powered Detection & RAG Pipeline

38 Disease Classes — PlantVillage Dataset

This document provides structured disease descriptions for all 38 classes in the PlantVillage/New Plant Diseases Dataset. Each entry includes pathogen type, causes, symptoms, treatment, and prevention — formatted specifically for use in Retrieval-Augmented Generation (RAG) pipelines for agricultural AI systems.

Crops covered: Apple, Blueberry, Cherry, Corn/Maize, Grape, Orange, Peach, Bell Pepper, Potato, Raspberry, Soybean, Squash, Strawberry, Tomato

Designed for: Nepal Agricultural AI Research | DenseNet121 + RAG System

Table of Contents

#	Disease Name	Plant	Pathogen Type
1	Apple Scab	Apple	Fungal
2	Apple Black Rot	Apple	Fungal
3	Apple Cedar Rust	Apple	Fungal
4	Apple Healthy	Apple	None
5	Blueberry Healthy	Blueberry	None
6	Cherry Powdery Mildew	Cherry	Fungal
7	Cherry Healthy	Cherry	None
8	Corn (Maize) Gray Leaf Spot	Corn / Maize	Fungal
9	Corn Common Rust	Corn / Maize	Fungal
10	Corn Northern Leaf Blight	Corn / Maize	Fungal
11	Corn Healthy	Corn / Maize	None
12	Grape Black Rot	Grape	Fungal
13	Grape Esca (Black Measles)	Grape	Fungal
14	Grape Leaf Blight (Isariopsis Leaf Spot)	Grape	Fungal
15	Grape Healthy	Grape	None
16	Orange Haunglongbing (Citrus Greening)	Orange / Citrus	Bacterial
17	Peach Bacterial Spot	Peach	Bacterial
18	Peach Healthy	Peach	None
19	Bell Pepper Bacterial Spot	Bell Pepper	Bacterial
20	Bell Pepper Healthy	Bell Pepper	None
21	Potato Early Blight	Potato	Fungal
22	Potato Late Blight	Potato	Oomycete (Water Mold)
23	Potato Healthy	Potato	None
24	Raspberry Healthy	Raspberry	None
25	Soybean Healthy	Soybean	None
26	Squash Powdery Mildew	Squash	Fungal
27	Strawberry Leaf Scorch	Strawberry	Fungal
28	Strawberry Healthy	Strawberry	None
29	Tomato Bacterial Spot	Tomato	Bacterial
30	Tomato Early Blight	Tomato	Fungal
31	Tomato Late Blight	Tomato	Oomycete (Water Mold)
32	Tomato Leaf Mold	Tomato	Fungal
33	Tomato Septoria Leaf Spot	Tomato	Fungal
34	Tomato Spider Mite (Two-spotted Spider Mite)	Tomato	Arthropod Pest
35	Tomato Target Spot	Tomato	Fungal
36	Tomato Yellow Leaf Curl Virus	Tomato	Viral
37	Tomato Mosaic Virus	Tomato	Viral

38	Tomato Healthy	Tomato	None
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1. Apple Scab

Plant: Apple **Pathogen Type:** Fungal **Pathogen:** *Venturia inaequalis*

Cause

Caused by the fungal pathogen *Venturia inaequalis*. Spreads via ascospores during wet spring weather. Overwinters in fallen infected leaves. Warm, moist conditions (18-24°C) with prolonged leaf wetness trigger infection.

Symptoms

- Olive-green to brown velvety spots on upper leaf surface
- Yellow halos around lesions on lower leaf surface
- Scabby, cracked lesions on fruit surface
- Premature defoliation in severe cases
- Distorted, stunted fruit growth

Treatment

- Apply fungicides: captan, myclobutanil, or mancozeb at first sign of infection
- Remove and destroy infected leaves and fruit
- Prune trees to improve air circulation
- Apply lime sulfur sprays during dormancy

Prevention

- Plant resistant apple varieties (e.g., Liberty, Redfree)
 - Rake and destroy fallen leaves in autumn
 - Avoid overhead irrigation
 - Apply preventive fungicide sprays before rain events during spring
 - Ensure proper orchard spacing for air circulation
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2. Apple Black Rot

Plant: Apple **Pathogen Type:** Fungal **Pathogen:** *Botryosphaeria obtusa*

Cause

Caused by the fungal pathogen *Botryosphaeria obtusa*. Fungus overwinters in dead bark, cankers, and mummified fruit. Warm, wet weather (24-29°C) favors infection. Spreads via conidia dispersed by rain.

Symptoms

- Circular brown spots with purple margins on leaves (frog-eye leaf spot)
- Fruit rot starting at the calyx end, turning black
- Concentric rings of black pycnidia on rotting fruit
- Cankers on twigs and branches with reddish-brown discoloration
- Mummified black fruit remaining on tree

Treatment

- Apply fungicides: captan or thiophanate-methyl
- Remove mummified fruit from trees and ground

- Prune out dead and cankered wood
- Disinfect pruning tools between cuts

Prevention

- Maintain tree vigor through proper fertilization and irrigation
 - Remove fire blight strikes promptly
 - Avoid wounding trees during cultivation
 - Apply dormant copper sprays to reduce inoculum
 - Keep orchards free of dead wood
-

3. Apple Cedar Rust

Plant: Apple **Pathogen Type:** Fungal **Pathogen:** *Gymnosporangium juniperi-virginianae*

Cause

Requires two host plants (juniper/cedar and apple) to complete its life cycle. Teliospores on cedar produce basidiospores that infect apple. Wet spring conditions (16-24°C) are required for spore germination and infection.

Symptoms

- Bright orange-yellow spots on upper surface of apple leaves
- Spore tubes (aecia) on lower leaf surface
- Orange lesions on fruit and fruit stems
- Swollen, distorted twigs
- Premature leaf drop in heavy infections

Treatment

- Apply fungicides: myclobutanil, propiconazole, or triadimefon
- Begin sprays when orange galls appear on nearby junipers
- Continue at 7-10 day intervals during wet periods

Prevention

- Remove nearby eastern red cedar and juniper trees if possible
 - Plant resistant apple varieties
 - Apply preventive fungicide sprays from pink bud stage
 - Avoid planting apple near juniper hosts
-

4. Apple Healthy

Plant: Apple **Pathogen Type:** None **Pathogen:** N/A

Cause

No disease present. Healthy apple leaf showing normal physiology.

Symptoms

- Deep green, uniform leaf color

- No spots, lesions, or discoloration
- Normal leaf shape and size

Treatment

- No treatment required

Prevention

- Continue regular orchard management
 - Monitor regularly for early disease signs
 - Maintain balanced fertilization and irrigation
-

5. Blueberry Healthy

Plant: Blueberry **Pathogen Type:** None **Pathogen:** N/A

Cause

No disease present. Healthy blueberry leaf showing normal physiology.

Symptoms

- Uniform green foliage
- No lesions or spots
- Normal growth pattern

Treatment

- No treatment required

Prevention

- Maintain soil pH 4.5-5.5
 - Avoid waterlogging
 - Monitor for pests and diseases regularly
-

6. Cherry Powdery Mildew

Plant: Cherry **Pathogen Type:** Fungal **Pathogen:** *Podosphaera clandestina*

Cause

Caused by obligate fungal pathogen *Podosphaera clandestina*. Unlike most fungi, favors dry conditions with moderate humidity (50-70%) and temperatures of 20-27°C. Spreads via airborne conidia. Overwinters as chasmothecia in bark crevices.

Symptoms

- White powdery coating on young leaves, shoots, and fruit
- Leaf curling and distortion
- Stunted shoot growth
- Brown scarring on fruit surface
- Premature leaf drop in severe infections

Treatment

- Apply sulfur-based fungicides or potassium bicarbonate
- Use systemic fungicides: myclobutanil or trifloxystrobin
- Remove heavily infected shoots
- Improve air circulation by pruning

Prevention

- Plant resistant cherry varieties
 - Avoid excessive nitrogen fertilization (promotes succulent growth)
 - Ensure good air circulation
 - Apply preventive fungicides during high-risk periods
-

7. Cherry Healthy

Plant: Cherry **Pathogen Type:** None **Pathogen:** N/A

Cause

No disease present.

Symptoms

- Glossy, dark green leaves
- No white coating or lesions
- Normal growth

Treatment

- No treatment required

Prevention

- Regular monitoring
 - Balanced nutrition
 - Proper pruning for air circulation
-

8. Corn (Maize) Gray Leaf Spot

Plant: Corn / Maize **Pathogen Type:** Fungal **Pathogen:** *Cercospora zeae-maydis*

Cause

Caused by *Cercospora zeae-maydis*. Overwinters in infected crop residue. High humidity (>90%), warm temperatures (25-30°C), and prolonged dew periods favor disease development. Spreads via airborne conidia.

Symptoms

- Rectangular, tan to gray lesions with parallel sides running along leaf veins
- Lesions appear water-soaked initially then turn gray-brown
- Premature death of lower leaves
- Significant yield loss in severe cases
- Lesions may coalesce to blight entire leaf

Treatment

- Apply strobilurin or triazole fungicides at first sign of disease
- Foliar sprays at tasseling to silking stage are most effective
- Remove and destroy infected crop residue after harvest

Prevention

- Plant resistant corn hybrids
 - Rotate crops (avoid continuous corn)
 - Till soil to bury infected residue
 - Avoid fields with history of disease
 - Ensure adequate plant spacing
-

9. Corn Common Rust

Plant: Corn / Maize **Pathogen Type:** Fungal **Pathogen:** *Puccinia sorghi*

Cause

Caused by obligate parasite *Puccinia sorghi*. Spreads via urediniospores carried by wind over long distances. Cool temperatures (16-23°C) with high humidity and dew favor infection. Does not overwinter in cold climates.

Symptoms

- Oval to elongated, powdery, brick-red to brown pustules on both leaf surfaces
- Pustules surrounded by yellow halo
- Heavily infected leaves turn yellow and die
- Pustules may appear on husks and stalks in severe cases
- Black telia (teliospores) form late in season

Treatment

- Apply fungicides: azoxystrobin, propiconazole, or mancozeb
- Early application at first pustule appearance is most effective
- Multiple applications may be needed in severe cases

Prevention

- Plant rust-resistant corn hybrids
 - Early planting to avoid peak infection periods
 - Monitor fields regularly, especially in cool, humid weather
 - Avoid overhead irrigation
-

10. Corn Northern Leaf Blight

Plant: Corn / Maize **Pathogen Type:** Fungal **Pathogen:** *Exserohilum turcicum*

Cause

Caused by *Exserohilum turcicum* (formerly *Helminthosporium turcicum*). Overwinters in infected corn residue. Moderate temperatures (18-27°C) and high humidity favor infection. Spreads by airborne conidia during wet

weather.

Symptoms

- Long, elliptical, grayish-green to tan lesions (2.5-15 cm) on leaves
- Lesions have wavy or irregular margins
- Dark green, water-soaked margins early in infection
- Lesions may coalesce to kill entire leaves
- Premature plant death in severe cases

Treatment

- Apply fungicides: azoxystrobin + propiconazole or pyraclostrobin
- Apply at early tassel stage for best results
- Destroy infected crop debris after harvest

Prevention

- Plant resistant or tolerant hybrids
- Crop rotation away from corn
- Tillage to decompose infected residue
- Avoid dense planting that reduces air circulation

11. Corn Healthy

Plant: Corn / Maize **Pathogen Type:** None **Pathogen:** N/A

Cause

No disease present.

Symptoms

- Uniform green, upright leaves
- No lesions, pustules, or blighting
- Normal growth and development

Treatment

- No treatment required

Prevention

- Crop rotation
- Balanced fertilization
- Regular field scouting

12. Grape Black Rot

Plant: Grape **Pathogen Type:** Fungal **Pathogen:** *Guignardia bidwellii*

Cause

Caused by *Guignardia bidwellii*. Overwinters in mummified berries and infected canes. Warm (21-30°C), wet weather during flowering and fruit development triggers infection. Spreads via splash-dispersed conidia.

Symptoms

- Small, reddish-brown circular spots with dark margins on leaves
- Black, shriveled (mummified) berries
- Tan lesions with black borders on berries initially
- Black pycnidia (spore bodies) visible in lesions
- Infected shoots and tendrils show dark, elongated lesions

Treatment

- Apply fungicides: myclobutanil, mancozeb, or captan
- Begin sprays at 2-3 cm shoot growth and continue through fruit development
- Remove and destroy mummified berries and infected canes
- Improve canopy ventilation by pruning

Prevention

- Remove all mummified fruit and infected plant material
- Train vines to improve air circulation
- Apply dormant copper sprays
- Avoid overhead irrigation
- Plant in well-drained locations

13. Grape Esca (Black Measles)

Plant: Grape **Pathogen Type:** Fungal **Pathogen:** *Phaeomoniella chlamydospora*, *Phaeoacremonium spp.*, *Fomitiporia mediterranea*

Cause

Complex disease caused by multiple fungal pathogens infecting woody tissues. Pathogens enter through pruning wounds. Stressful conditions (drought, high temperatures) and wetting/drying cycles trigger symptom expression.

Symptoms

- Tiger-stripe pattern (interveinal chlorosis and necrosis) on leaves
- Small, dark spots on berries (measles symptoms)
- Internal wood discoloration (black or brown)
- Sudden wilting and death of shoots (apoplexy) in hot weather
- Progressive decline of vine health over years

Treatment

- No curative treatment available once wood is infected
- Remove and destroy severely infected vines
- Remedial surgery: remove infected wood back to healthy tissue
- Trunk renewal by training new canes

Prevention

- Protect pruning wounds with fungicide paste (thiophanate-methyl) or sealant
- Prune during dry weather
- Avoid large wounds
- Use certified disease-free planting material
- Maintain vine health with proper irrigation and fertilization

14. Grape Leaf Blight (Isariopsis Leaf Spot)

Plant: Grape **Pathogen Type:** Fungal **Pathogen:** *Pseudocercospora vitis*

Cause

Caused by *Pseudocercospora vitis*. Favored by warm, humid conditions. Spreads via conidia during wet periods. More common in tropical and subtropical regions.

Symptoms

- Irregular dark brown spots on upper leaf surface
- Grayish sporulation on lower leaf surface
- Premature defoliation
- Reduction in photosynthetic capacity
- Weakening of vine overall

Treatment

- Apply copper-based fungicides or mancozeb
- Remove affected leaves
- Improve canopy management

Prevention

- Adequate vine spacing for air circulation
- Avoid overhead irrigation
- Remove fallen leaves
- Regular preventive fungicide applications in high-risk areas

15. Grape Healthy

Plant: Grape **Pathogen Type:** None **Pathogen:** N/A

Cause

No disease present.

Symptoms

- Uniform green leaves
- No spots or lesions
- Healthy fruit development

Treatment

- No treatment required

Prevention

- Regular scouting
- Good canopy management
- Balanced irrigation and fertilization

16. Orange Haunglongbing (Citrus Greening)

Plant: Orange / Citrus **Pathogen Type:** Bacterial **Pathogen:** *Candidatus Liberibacter asiaticus*

Cause

Caused by the bacterium *Candidatus Liberibacter asiaticus*, transmitted by the Asian citrus psyllid (*Diaphorina citri*). Insect vector feeds on phloem and transmits bacteria. Once infected, the tree cannot be cured.

Symptoms

- Asymmetric (blotchy) yellow mottling of leaves (huanglongbing means yellow dragon disease)
- Small, lopsided, bitter fruit with green coloration at styler end
- Twig dieback and poor flowering
- Stunted growth and premature fruit drop
- Zinc deficiency-like symptoms (corky veins)

Treatment

- No curative treatment exists
- Remove and destroy infected trees to prevent spread
- Inject antibiotics (oxytetracycline) to temporarily suppress symptoms — not a cure
- Control psyllid vector aggressively with insecticides

Prevention

- Use certified disease-free nursery stock
- Control Asian citrus psyllid populations with systemic insecticides
- Establish psyllid-proof nurseries
- Quarantine new plant material
- Regular grove monitoring for psyllid and disease symptoms

17. Peach Bacterial Spot

Plant: Peach **Pathogen Type:** Bacterial **Pathogen:** *Xanthomonas arboricola pv. pruni*

Cause

Caused by *Xanthomonas arboricola pv. pruni*. Overwinters in infected twigs. Warm, wet, and windy conditions spread bacteria via rain splash and wind. Infections occur through stomata and wounds.

Symptoms

- Small, angular, water-soaked spots on leaves becoming brown with yellow halos
- Shot-hole appearance as infected tissue falls out
- Raised, rough spots on fruit (scab-like)

- Fruit cracking in severe cases
- Twig cankers causing dieback

Treatment

- Apply copper-based bactericides (copper hydroxide, copper sulfate)
- Apply at petal fall and continue at 10-14 day intervals
- Prune out infected twigs during dry weather

Prevention

- Plant resistant peach varieties
- Avoid overhead irrigation
- Proper orchard spacing for air circulation
- Apply copper sprays at leaf fall and dormant stages
- Avoid excessive nitrogen fertilization

18. Peach Healthy

Plant: Peach **Pathogen Type:** None **Pathogen:** N/A

Cause

No disease present.

Symptoms

- Uniform, glossy green leaves
- No spots or lesions
- Normal fruit development

Treatment

- No treatment required

Prevention

- Regular monitoring
- Copper dormant sprays as preventive measure
- Good pruning practices

19. Bell Pepper Bacterial Spot

Plant: Bell Pepper **Pathogen Type:** Bacterial **Pathogen:** *Xanthomonas euvesicatoria*

Cause

Caused by *Xanthomonas euvesicatoria*. Spread by rain splash, wind-driven rain, and contaminated seed. Warm temperatures (24-30°C) and high humidity favor rapid spread. Seed is the primary long-distance carrier.

Symptoms

- Small, water-soaked spots on leaves that enlarge and turn brown
- Yellow halos surrounding dark lesions

- Shot-hole appearance as lesions dry and fall out
- Raised, brown, scabby spots on fruit
- Defoliation under severe infection

Treatment

- Apply copper bactericides (copper hydroxide) mixed with mancozeb
- Begin at first sign of disease and repeat every 5-7 days during wet weather
- Remove heavily infected plants

Prevention

- Use certified disease-free or treated seed
- Avoid overhead irrigation
- Crop rotation (avoid peppers and tomatoes in same location)
- Disinfect tools and equipment
- Plant in well-drained areas with good air movement

20. Bell Pepper Healthy

Plant: Bell Pepper **Pathogen Type:** None **Pathogen:** N/A

Cause

No disease present.

Symptoms

- Bright green, smooth leaves
- No spots or wilting
- Normal plant structure

Treatment

- No treatment required

Prevention

- Regular monitoring
- Drip irrigation preferred
- Crop rotation

21. Potato Early Blight

Plant: Potato **Pathogen Type:** Fungal **Pathogen:** *Alternaria solani*

Cause

Caused by *Alternaria solani*. Overwinters in infected plant debris. Warm temperatures (24-29°C) and alternating wet and dry periods favor sporulation and infection. Older, stressed leaves are most susceptible.

Symptoms

- Dark brown, circular to oval lesions with concentric rings (target-board pattern)

- Yellow halo surrounding lesions
- Lesions appear first on older, lower leaves
- Severe defoliation starting from bottom of plant
- Dark, sunken lesions at stem-end of tubers

Treatment

- Apply fungicides: chlorothalonil, mancozeb, or azoxystrobin
- Begin at first sign of infection or at-risk periods
- Remove and destroy infected plant material
- Ensure adequate potassium fertilization

Prevention

- Use certified disease-free seed potatoes
- Crop rotation (3-4 year cycle away from potatoes and tomatoes)
- Avoid overhead irrigation or water in morning
- Maintain plant vigor with balanced fertilization
- Hill soil around plants to reduce tuber infection

22. Potato Late Blight

Plant: Potato **Pathogen Type:** Oomycete (Water Mold) **Pathogen:** *Phytophthora infestans*

Cause

Caused by oomycete *Phytophthora infestans* — responsible for the Irish Potato Famine (1845). Spreads rapidly under cool (10-20°C), wet, and humid conditions. Zoospores spread via wind and rain splash. Can devastate a crop within days under favorable conditions.

Symptoms

- Water-soaked, pale green lesions on leaf margins turning dark brown/black
- White fluffy sporulation on underside of leaves in humid conditions
- Rapid collapse and blackening of entire plant
- Reddish-brown, firm rot in tuber flesh
- Foul smell from infected tubers

Treatment

- Apply fungicides: metalaxyl + mancozeb, cymoxanil, or fluopicolide
- Begin preventive applications when conditions favor disease
- Apply every 5-7 days during cool, wet weather
- Remove and destroy infected plant material — do not compost
- Harvest tubers before disease reaches them

Prevention

- Use certified disease-free seed potatoes
- Plant resistant varieties (e.g., Sarpo Mira, Defender)
- Avoid overhead irrigation
- Hill soil to protect tubers from spore wash-down
- Crop rotation and removal of volunteer potato plants

- Monitor using Blight forecast models (e.g., BlightCast)

23. Potato Healthy

Plant: Potato **Pathogen Type:** None **Pathogen:** N/A

Cause

No disease present.

Symptoms

- Dark green, turgid leaves
- No lesions or wilting
- Healthy stem and canopy

Treatment

- No treatment required

Prevention

- Certified seed potatoes
- Crop rotation
- Regular field scouting

24. Raspberry Healthy

Plant: Raspberry **Pathogen Type:** None **Pathogen:** N/A

Cause

No disease present.

Symptoms

- Bright green, typical leaf shape
- No spots or lesions
- Normal cane growth

Treatment

- No treatment required

Prevention

- Monitor for cane blight and botrytis
- Good air circulation
- Avoid waterlogging

25. Soybean Healthy

Plant: Soybean **Pathogen Type:** None **Pathogen:** N/A

Cause

No disease present.

Symptoms

- Uniform trifoliolate leaves
- No yellowing or spots
- Normal growth

Treatment

- No treatment required

Prevention

- Crop rotation
- Balanced nutrition
- Regular scouting

26. Squash Powdery Mildew

Plant: Squash **Pathogen Type:** Fungal **Pathogen:** *Podosphaera xanthii* / *Erysiphe cichoracearum*

Cause

Caused by *Podosphaera xanthii* primarily. Unlike most fungi, powdery mildew thrives in dry conditions with moderate humidity. Warm days (20-28°C) and cool nights favor disease. Spreads via airborne conidia. Does not require free water for infection.

Symptoms

- White, powdery fungal growth on upper and lower leaf surfaces
- Yellow patches beneath white growth
- Severely infected leaves turn brown and die
- Stunted plant growth
- Reduced fruit size and quality

Treatment

- Apply sulfur-based fungicides or potassium bicarbonate (organic option)
- Apply neem oil as an organic management option
- Use systemic fungicides: trifloxystrobin or myclobutanil
- Remove and destroy heavily infected leaves

Prevention

- Plant resistant squash varieties
 - Ensure good air circulation with adequate spacing
 - Avoid excessive nitrogen fertilization
 - Apply preventive fungicide sprays in high-risk conditions
 - Water at soil level — avoid wetting foliage
-

27. Strawberry Leaf Scorch

Plant: Strawberry **Pathogen Type:** Fungal **Pathogen:** *Diplocarpon earliana*

Cause

Caused by *Diplocarpon earliana*. Favored by warm (20-30°C), wet weather. Spreads via ascospores and conidia during wet conditions. Overwinters in infected leaf tissue.

Symptoms

- Numerous small, irregular, dark purple to reddish-purple spots on upper leaf surface
- Spots may coalesce giving scorched appearance
- Tan to gray centers in older lesions
- Leaf margins turn brown and curl upward
- Severe defoliation weakens plants

Treatment

- Apply fungicides: captan, myclobutanil, or azoxystrobin
- Begin applications at first sign of disease
- Remove infected leaves and plant debris

Prevention

- Plant certified disease-free transplants
 - Remove old leaves after harvest
 - Avoid overhead irrigation
 - Plant in well-drained locations
 - Plant resistant strawberry varieties
-

28. Strawberry Healthy

Plant: Strawberry **Pathogen Type:** None **Pathogen:** N/A

Cause

No disease present.

Symptoms

- Bright green leaves
- No spots or scorching
- Normal runner production

Treatment

- No treatment required

Prevention

- Regular monitoring
 - Avoid waterlogging
 - Remove old foliage after harvest
-

29. Tomato Bacterial Spot

Plant: Tomato **Pathogen Type:** Bacterial **Pathogen:** *Xanthomonas perforans*

Cause

Caused by *Xanthomonas perforans*. Spreads by rain splash, contaminated seed, and transplants. Warm (24-30°C) and wet conditions favor rapid spread. Seed is primary long-distance carrier.

Symptoms

- Small, brown, water-soaked spots on leaves with yellow halos
- Shot-hole as infected tissue falls away
- Small, raised, scabby spots on fruit
- Severe defoliation in wet conditions
- Blossom drop reducing yield

Treatment

- Apply copper bactericide + mancozeb mixture
- Spray every 5-7 days during wet weather
- Remove and destroy infected plant material

Prevention

- Use certified disease-free or hot water-treated seed
 - Avoid overhead irrigation
 - Crop rotation (3-4 years)
 - Disinfect tools and stakes
 - Plant resistant tomato varieties
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30. Tomato Early Blight

Plant: Tomato **Pathogen Type:** Fungal **Pathogen:** *Alternaria solani*

Cause

Caused by *Alternaria solani*. Overwinters in infected debris. Warm (24-29°C) and alternating wet/dry periods favor sporulation. Older, stressed leaves are first affected. Same pathogen as Potato Early Blight.

Symptoms

- Target-like circular lesions with concentric rings on older leaves
- Yellow halo around lesions
- Disease progresses from lower to upper canopy
- Severe defoliation
- Sunken, dark lesions at stem end of fruit

Treatment

- Apply fungicides: chlorothalonil, azoxystrobin, or copper-based products
- Remove infected leaves promptly
- Stake plants to improve air circulation

Prevention

- Crop rotation (2-3 years away from tomatoes/potatoes)
 - Use certified, disease-free transplants
 - Stake and prune for air circulation
 - Mulch to prevent soil splash
 - Drip irrigation preferred
-

31. Tomato Late Blight

Plant: Tomato **Pathogen Type:** Oomycete (Water Mold) **Pathogen:** *Phytophthora infestans*

Cause

Caused by the same pathogen as Potato Late Blight. Cool (10-20°C) and wet conditions favor rapid spread. Can destroy a planting in days under ideal conditions. Spreads easily from potato crops nearby.

Symptoms

- Greasy, water-soaked, pale green to brown lesions on leaves
- White sporulation on underside of lesions in high humidity
- Rapid browning and collapse of foliage
- Dark brown, firm lesions on stems and fruit
- Entire plant collapse in advanced cases

Treatment

- Apply fungicides immediately: metalaxyl, cymoxanil, or fluopicolide
- Preventive applications before wet/cool periods are critical
- Remove and bag infected material — do not compost
- Apply every 5-7 days in risk conditions

Prevention

- Use resistant tomato varieties
 - Avoid planting near potatoes
 - Avoid overhead irrigation
 - Remove volunteer potato and tomato plants
 - Use forecasting tools for spray timing
-

32. Tomato Leaf Mold

Plant: Tomato **Pathogen Type:** Fungal **Pathogen:** *Passalora fulva* (syn. *Fulvia fulva*)

Cause

Caused by *Passalora fulva*. Primarily a problem in greenhouse-grown tomatoes. High humidity (>85%) and mild temperatures (22-25°C) favor disease. Spreads via airborne conidia.

Symptoms

- Yellow patches on upper leaf surface

- Olive-green to gray velvety mold growth on lower leaf surface
- Infected leaves curl upward and turn brown
- Severe infection causes defoliation
- Rarely affects fruit directly

Treatment

- Reduce humidity by improving greenhouse ventilation
- Apply fungicides: chlorothalonil, mancozeb, or azoxystrobin
- Remove and destroy infected leaves

Prevention

- Maintain humidity below 85% in greenhouses
 - Ensure adequate plant spacing
 - Use resistant tomato varieties (Cf genes)
 - Avoid wetting foliage during irrigation
 - Sanitize greenhouse structures between crops
-

33. Tomato Septoria Leaf Spot

Plant: Tomato **Pathogen Type:** Fungal **Pathogen:** *Septoria lycopersici*

Cause

Caused by *Septoria lycopersici*. Overwinters in infected debris and on solanaceous weeds. Warm (20-25°C), wet conditions favor rapid spread. Spreads by rain splash and contact. First affects lower leaves after fruit set.

Symptoms

- Numerous small, circular spots with white/gray centers and dark brown margins
- Tiny black pycnidia (spore bodies) visible in lesion centers
- Lesions appear first on lower leaves and progress upward
- Severe defoliation leading to sun-scalded fruit
- Stems and calyxes may also be infected

Treatment

- Apply fungicides: chlorothalonil, copper-based, or mancozeb
- Begin applications when first lesions appear
- Remove infected lower leaves
- Repeat every 7-10 days during wet weather

Prevention

- Crop rotation (3+ years away from tomatoes)
 - Stake plants and prune suckers for air circulation
 - Mulch to prevent soil splash
 - Remove solanaceous weeds from field borders
 - Use drip irrigation
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34. Tomato Spider Mite (Two-spotted Spider Mite)

Plant: Tomato **Pathogen Type:** Arthropod Pest **Pathogen:** *Tetranychus urticae*

Cause

Caused by the two-spotted spider mite, *Tetranychus urticae*. Hot, dry weather (above 27°C) and dusty conditions favor rapid population build-up. Mites pierce leaf cells to feed on contents. Can complete a generation in under 7 days in hot weather.

Symptoms

- Tiny yellow or white stippling dots on upper leaf surface
- Fine webbing on undersides of leaves in heavy infestations
- Leaves turn bronze, brown, and dry out
- Severe leaf drop and plant stress
- Tiny moving dots (mites) visible with hand lens on undersides of leaves

Treatment

- Apply acaricides: abamectin, bifenazate, or spiromesifen
- Apply insecticidal soap or neem oil for organic management
- Introduce predatory mites (*Phytoseiulus persimilis*) for biological control
- Strong water sprays to dislodge mites

Prevention

- Avoid drought stress in plants
- Avoid dusty conditions (control road dust near fields)
- Avoid broad-spectrum insecticide use that kills natural enemies
- Regular monitoring with hand lens
- Maintain adequate soil moisture

35. Tomato Target Spot

Plant: Tomato **Pathogen Type:** Fungal **Pathogen:** *Corynespora cassiicola*

Cause

Caused by *Corynespora cassiicola*, a widespread fungal pathogen. Warm (25-32°C), humid conditions favor disease. Spreads via airborne conidia. Can affect fruit, stems, and leaves.

Symptoms

- Brown, circular lesions with concentric rings on leaves
- Lesions have yellow margins and may have a target-like appearance
- Dark, sunken spots on fruit
- Stem lesions causing plant wilting
- Defoliation in severe cases

Treatment

- Apply fungicides: chlorothalonil, azoxystrobin, or mancozeb
- Remove and destroy infected leaves

- Improve air circulation

Prevention

- Stake and train plants for air circulation
- Avoid overhead irrigation
- Crop rotation
- Mulch soil to prevent splash dispersal

36. Tomato Yellow Leaf Curl Virus

Plant: Tomato **Pathogen Type:** Viral **Pathogen:** *Tomato Yellow Leaf Curl Virus (TYLCV)*

Cause

Caused by TYLCV, transmitted by the silverleaf whitefly (*Bemisia tabaci*) in a persistent, circulative manner. Whiteflies acquire virus in 15-30 minutes and can transmit for life. No seed transmission. Warm conditions favor whitefly populations.

Symptoms

- Severe stunting of new growth
- Upward curling and yellowing (chlorosis) of leaf margins
- Leaves become small, crumpled, and cup-shaped
- Flower drop and very low fruit set
- Plants infected early produce no marketable fruit

Treatment

- No cure once infected — remove and destroy infected plants
- Control whitefly vector with imidacloprid, thiamethoxam, or insect growth regulators
- Apply reflective mulches to deter whiteflies
- Use yellow sticky traps for monitoring and mass trapping

Prevention

- Plant resistant tomato varieties (Ty genes)
- Use insect-proof net tunnels for seedling production
- Apply systemic insecticide at transplanting
- Rogue out and destroy infected plants immediately
- Maintain weed-free borders
- Avoid planting near old infected tomato fields

37. Tomato Mosaic Virus

Plant: Tomato **Pathogen Type:** Viral **Pathogen:** *Tomato Mosaic Virus (ToMV)*

Cause

Caused by ToMV, a tobamovirus. Extremely stable and persistent in soil, seed, and on tools. Spreads primarily by mechanical contact — hands, tools, clothing. Seed-borne. Does NOT require an insect vector.

Symptoms

- Light and dark green mosaic or mottling pattern on leaves
- Leaf distortion, curling, and blistering
- Stunted plant growth
- Fruit may show yellow blotches or internal browning
- Reduced fruit set and quality

Treatment

- No chemical cure available
- Remove and destroy infected plants
- Disinfect tools with 10% bleach or trisodium phosphate solution
- Wash hands thoroughly before handling plants

Prevention

- Use certified virus-free or tested seed
- Plant resistant varieties (Tm genes)
- Disinfect tools, stakes, and hands during all operations
- Remove and destroy infected plants and debris
- Do not smoke near plants (tobacco can carry ToMV)
- Soil solarization to reduce soil-borne virus

38. Tomato Healthy

Plant: Tomato **Pathogen Type:** None **Pathogen:** N/A

Cause

No disease present.

Symptoms

- Dark green, turgid leaves
- No spots, lesions, or distortion
- Normal growth and fruit development

Treatment

- No treatment required

Prevention

- Regular field scouting
 - Good cultural practices
 - Crop rotation
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